

Summative Report: Predicting Medical Insurance Charges

Oct 3, 2026 · @nothing

Results

In this project I tried to predict how much people pay for medical insurance in a year, so my target variable was charges. I used the insurance dataset from Lantz (2013), which has information like age, BMI, number of children, sex, smoking and region. First I cleaned the data by removing one duplicate row and changing the text columns into numbers, and then I used 80% of the data to train the model and 20% to test it. My first multiple linear regression model got an R^2 of 0.81 and an RMSE of about \$5,956, and when I tried using the log of charges it actually got worse ($R^2 = 0.72$). In my scatter plots I noticed that smokers with a BMI over 30 paid a lot more, so I added this to the model along with age squared. This made the model much better, with an R^2 of 0.91 and an RMSE of about \$4,158, which means it explains around 91% of the differences in charges. The biggest finding was that smoking matters the most, and smokers who are also obese pay about \$19,600 more per year on top of what smoking already adds. Older people also pay more, but sex and region did not really make a difference. The residual plots showed the model works well for most people, but it still underestimates a few people with very high charges, probably because of health problems that are not in the data.

References

Lantz, B. (2013). *Machine learning with R*. Packt Publishing.

Stedy. (n.d.). *Machine-Learning-with-R-datasets* [Data set]. GitHub. Retrieved October 3, 2026, from <https://github.com/stedy/Machine-Learning-with-R-datasets>

AI use statement

Tool used: Claude (Anthropic), on October 3, 2026.

Prompts: I asked Claude to explain the assignment, fix a file-loading error, assess my first dataset (Instagram comments), suggest a better dataset with references, and help me complete the proposal, analysis notebook, and report within my time limit.

Output: Claude explained the assignment, showed that the Instagram data could not support a useful regression model, downloaded the Medical Cost Personal Dataset, drafted the proposal and this report, and wrote the Python notebook for cleaning, visualization, and regression. [Paste the full conversation or its key prompts and outputs here.]

How I validated the content: [Write 5–10 sentences in your own words. For example: I ran the notebook myself and confirmed that every number in the report (R^2 , RMSE, coefficients) matched my own output. I checked the dataset source on GitHub and Kaggle. I reviewed each plot and confirmed the written interpretations matched what I saw. I compared the methods with the course material on regression.]